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GenAI-Assisted Anti-Money Laundering (AML) Compliance System

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CERTIFICATE

This is to certify that the dissertation entitled **"GENAI-ASSISTED ANTI-MONEY LAUNDERING (AML) COMPLIANCE SYSTEM"** being submitted by A. VINAY (23RA1A66D0), D. MALLESH (24RA5A6603), CH. SURYA TEJA (23RA1A66C1), T. ROHITH SINGH (23RA1A6684) in partial fulfilment for the award of Bachelor of Technology in Computer Science and Engineering (AI&ML) to the Kommuri Pratap Reddy Institute of Technology is a record of confined work carried out by them under my guidance and supervision.

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DECLARATION

We, A. VINAY (23RA1A66D0), D. MALLESH (24RA5A6603), CH. SURYA TEJA (23RA1A66C1), T. ROHITH SINGH (23RA1A6684) here by declare that the project report titled "**GENAI-ASSISTED ANTI-MONEY LAUNDERING (AML) COMPLIANCE SYSTEM**" under the guidance of SANTHI KUMARI, Assistant Professor of Computer Science and Engineering (AI&ML), Kommuri Pratap Reddy Institute of Technology, Ghanpur, is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI&ML). This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this thesis have not been submitted to any other University and Institute for the award of any Degree or Diploma.

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To emerge as a premier institute for high quality professional graduates who can contribute to economic and social developments of the Nation.

Mission of the Institute:

Mission	Statement
IM1	To have holistic approach in curriculum and pedagogy through industry interface to meet the needs of Global Competency.
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PSO3	Use tools and techniques to solve problems in AI & ML.

GenAI-Assisted AML Compliance System

ABSTRACT

Money laundering is a major financial crime that threatens the integrity of financial institutions and the global economy. Conventional Anti-Money Laundering (AML) systems mainly rely on predefined rules, fixed thresholds, and manual investigation. These approaches often generate excessive false positives, require considerable human effort, and may fail to detect complex or previously unseen laundering patterns.

This project proposes a GenAI-Assisted AML Compliance System that combines Machine Learning, Artificial Intelligence, Natural Language Processing, Generative AI, and explainable analytics to strengthen transaction monitoring and suspicious activity detection. The system analyzes transaction amount, frequency, location, account behavior, beneficiary relationships, and transaction history to identify anomalous patterns and assign customer or transaction risk scores. Generative AI assists compliance analysts by providing understandable reasons for alerts, summarizing investigation findings, identifying contributing risk factors, and supporting the preparation of compliance reports.

A secure investigation dashboard enables authorized personnel to review alerts, examine customer and transaction risk information, manage investigation status, and prioritize high-risk cases. The system also incorporates access controls, audit logging, data protection, explainability, and human validation to support responsible AI use and regulatory compliance. By reducing unnecessary alerts and improving investigation efficiency, the proposed solution provides a scalable and interpretable decision-support system while ensuring that final compliance decisions remain under human oversight.

Keywords: *Anti-Money Laundering, Generative AI, Machine Learning, Anomaly Detection, Risk Scoring, Explainable AI, Natural Language Processing, Transaction Monitoring, Audit Logging, Human Validation*

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CHAPTER 1

INTRODUCTION

1.1 Overview

Anti-Money Laundering (AML) is a critical financial compliance function intended to identify and investigate activities that may be associated with money laundering and related financial crime. Modern transaction environments generate large volumes of financial activity across accounts, beneficiaries, locations, channels, and time periods. The project abstract identifies that conventional AML monitoring is commonly driven by predefined rules, fixed thresholds, and manual investigation. Such approaches can create high alert volumes and may not reliably expose complex or previously unseen patterns. The proposed GenAI-Assisted AML Compliance System combines Machine Learning, Artificial Intelligence, Natural Language Processing, Generative AI, and explainable analytics. Its purpose is to strengthen transaction monitoring by examining transaction amount, frequency, location, account behavior, beneficiary relationships, and transaction history. These signals are used to identify anomalous behavior and support the assignment of customer or transaction risk scores. Generative AI is positioned as an analyst-assistance layer rather than as an autonomous compliance authority. It can explain why an alert was generated, summarize investigation findings, identify contributing risk factors, and support the preparation of compliance reports. A secure investigation dashboard provides authorized personnel with a consolidated view of alerts, customer and transaction risk information, and investigation status. The overall design emphasizes explainability, access controls, audit logging, data protection, and human validation. The final compliance decision remains under human oversight, making the proposed solution a decision-support platform rather than a replacement for qualified compliance professionals.

1.2 Problem Definition

Traditional AML monitoring faces a practical balance between sensitivity and investigation workload. Rule-based systems can identify known risk conditions, but fixed thresholds may generate large numbers of alerts and may not capture behavioral patterns that do not match predefined scenarios. Manual investigation can then become time-consuming, especially when analysts must examine transaction histories and related contextual information separately.

• **High false-positive workload:** Predefined rules and fixed thresholds can flag legitimate activity, increasing the number of alerts requiring analyst review.

• **Limited pattern discovery:** Previously unseen or complex laundering behavior may not match a predefined rule or threshold.

• **Fragmented investigation context:** Transaction, account, beneficiary, location, and historical information may need to be examined together to understand risk.

• **Difficult explanations:** An alert score alone may not provide an understandable narrative explaining the factors that contributed to the alert.

• **Manual reporting effort:** Compliance analysts may spend significant time summarizing findings and preparing investigation documentation.

• **Need for human accountability:** High-impact compliance decisions require controlled review, validation, and traceability rather than fully autonomous generation.

Formal Problem Statement: To design a secure, explainable, and scalable AML decision-support system that combines anomaly detection, risk scoring, and Generative AI-assisted investigation so that compliance analysts can prioritize suspicious cases, understand contributing risk factors, and prepare investigation reports while retaining human validation and regulatory oversight.

1.3 Research Motivation

The motivation for the project arises from the gap between the volume and complexity of financial transactions and the practical capacity of human investigation teams. The project abstract specifically identifies excessive false positives, substantial human effort, and difficulty detecting complex or previously unseen laundering patterns as important weaknesses of conventional approaches. Machine Learning and anomaly-oriented analytics can provide an additional layer for identifying unusual behavior, while Natural Language Processing and Generative AI can convert analytical findings into understandable summaries. Combining these capabilities can help an analyst move from an alert-centric workflow toward a risk-prioritized and explanation-supported investigation process. The project is also motivated by responsible AI requirements. Because AML outcomes can affect financial access and institutional risk decisions, the system is designed around access control, audit logging, data protection, explainability, and human validation. These controls are treated as core system characteristics rather than optional additions.

1.4 Research Objectives

The primary objectives of the proposed GenAI-Assisted AML Compliance System are:

1. • **1. Develop a transaction-monitoring framework:** Organize transaction, customer, beneficiary, location, and historical information into an investigation-oriented processing flow.
2. • **2. Detect anomalous behavior:** Use analytical and Machine Learning techniques to identify unusual transaction patterns and support risk assessment.
3. • **3. Generate risk scores:** Assign customer or transaction risk scores that help investigators prioritize cases for review.
4. • **4. Provide GenAI-assisted explanations:** Generate understandable reasons for alerts and identify the major contributing risk factors.
5. • **5. Summarize investigations:** Support analysts by converting investigation findings into concise, structured compliance-oriented narratives.
6. • **6. Provide a secure investigation dashboard:** Enable authorized personnel to review alerts, inspect risk information, and manage investigation status.
7. • **7. Maintain auditability:** Record appropriate investigation actions and audit information so that the review process remains traceable.
8. • **8. Protect sensitive information:** Incorporate access controls and data-protection practices appropriate for sensitive financial information.
9. • **9. Preserve human oversight:** Keep final compliance decisions under qualified human validation rather than delegating them completely to AI.
10. • **10. Support scalable decision assistance:** Provide an architecture that can handle growing transaction-monitoring and investigation workloads without making the final decision autonomous.

1.5 Significance

The project is significant because it combines multiple AI capabilities into a single compliance-support workflow. From a technical perspective, it demonstrates how anomaly detection, risk scoring, NLP, and Generative AI can work together. From an operational perspective, it aims to reduce unnecessary investigation effort and help analysts focus on higher-risk cases. From an explainability perspective, it adds natural-language reasoning and investigation summaries instead of presenting only an opaque alert score.

The project also highlights the importance of human-centered AI in financial compliance. The proposed system is intended to assist authorized personnel, not to independently determine whether a customer or transaction is legally involved in money laundering. This distinction is central to the design because explanations, scores, and alerts must remain reviewable and contestable by human investigators.

1.6 Applications

• **Transaction monitoring:** Screening customer transactions for unusual amount, frequency, location, or behavioral patterns.

• **Customer risk assessment:** Supporting risk scoring using account behavior and transaction history.

• **Suspicious activity investigation:** Helping analysts inspect alerts and understand contributing factors.

• **Beneficiary relationship analysis:** Highlighting relationships and transaction patterns involving beneficiaries.

• **Compliance reporting:** Assisting with the preparation of investigation summaries and compliance reports.

• **Case management support:** Allowing authorized investigators to review alerts and manage investigation status.

1.7 Advantages

• **Explainable alerts:** Provides understandable reasons and contributing factors for generated alerts.

• **Reduced investigation effort:** Automates parts of information summarization and report preparation.

• **Pattern-oriented monitoring:** Supports analysis beyond fixed thresholds by considering behavioral and historical signals.

• **Secure access:** Provides controlled access to sensitive investigation information.

• **Auditability:** Supports traceable investigation activity through audit logging.

• **Human validation:** Keeps final compliance decisions with authorized human reviewers.

• **Scalability:** Provides a foundation for handling large volumes of transaction-monitoring activity.

CHAPTER 2

LITERATURE SURVEY

2.1 Survey of Related Literature

The project is positioned at the intersection of AML transaction monitoring, Machine Learning-based anomaly detection, explainable Artificial Intelligence, Natural Language Processing, and Generative AI. Conventional AML literature and operational practice commonly emphasize rule-based monitoring, threshold conditions, customer risk indicators, and analyst investigation. These mechanisms remain important because they provide explicit and reviewable conditions for known risk scenarios. Machine Learning introduces a complementary approach in which patterns can be learned or identified from transaction and behavioral information. In an AML context, analytical variables can include transaction amount, frequency, location, account behavior, beneficiary relationships, and historical activity. Such features provide a broader behavioral view than a single fixed rule. Explainable AI focuses on making model outputs understandable to human users. For AML investigation, explainability is particularly useful because analysts need to understand why a transaction or customer was prioritized. Natural-language explanations can complement numerical risk scores by presenting contributing factors in a form that can be reviewed during investigation. Generative AI extends this capability by producing summaries and narrative explanations. In the proposed system, GenAI is not used as the sole detection mechanism. Instead, it assists the human analyst after analytical signals have been generated, helping explain alerts, summarize findings, identify risk factors, and support compliance report preparation.

2.2 Detailed Analytical Review

A conventional rule engine is strong at expressing explicit conditions, such as threshold violations or predefined transaction scenarios. Its limitation is that it is dependent on the quality and coverage of the rules. When behavior changes or when suspicious activity appears in a new form, the relevant rule may not exist. This creates a need for complementary analytical techniques. Machine Learning can examine relationships among multiple variables and identify patterns that may be difficult to encode manually. In the proposed system, ML-oriented analysis is associated with anomaly detection and risk scoring. However, model output alone is not sufficient for an investigation workflow because a numerical score does

not automatically communicate the evidence considered by the model. NLP and Generative AI provide a natural-language interface to analytical findings. They can transform structured alert information into readable explanations and investigation summaries. This is useful for analyst productivity, but generated text must be treated as an assistance mechanism because generative models can produce incorrect or incomplete statements. Therefore, the proposed architecture keeps human validation as a required control. The combination of analytical detection and explainable generation therefore addresses two different needs: identifying cases that deserve attention and helping investigators understand those cases. Security and governance controls then provide a third layer by controlling access, recording audit information, protecting sensitive data, and preserving accountability.

2.3 Research Gap Analysis

• **Fixed-rule dependence:** Conventional systems may depend heavily on predefined rules and thresholds, which can create high alert volumes and limited coverage of novel behavior.

• **Alert-to-explanation gap:** Risk scores or alerts do not necessarily provide an analyst-friendly narrative explaining the main contributing factors.

• **Investigation summarization gap:** Analysts may need to manually combine transaction history and alert information into a coherent investigation summary.

• **Human-AI integration gap:** An effective AML system must combine automated analysis with controlled human review rather than treating AI output as an unquestionable decision.

• **Governance gap:** Sensitive financial information requires explicit access control, audit logging, data protection, and traceability within the application.

Addressing the Gap: The proposed system brings these capabilities together in one workflow: transaction and behavioral information is analyzed for anomalies, risk is prioritized, Generative AI explains and summarizes the findings, an investigation dashboard presents the evidence to authorized personnel, and human validation remains responsible for final compliance decisions.

CHAPTER 3

EXISTING SYSTEM

3.1 Traditional AML Systems

Traditional AML monitoring commonly relies on a combination of predefined rules, fixed thresholds, transaction screening, customer risk indicators, and manual investigation. A monitoring engine checks incoming activity against configured conditions. When a condition is triggered, an alert is generated and routed to an analyst for review. Rule-based monitoring is valuable because its logic is explicit and relatively easy to document. Analysts can understand which condition triggered an alert. However, large financial transaction environments can generate many alerts, including legitimate activity that happens to satisfy a rule. The resulting false-positive workload can consume significant investigation capacity. Manual investigation remains necessary because an alert generally requires contextual review. Analysts may need to examine transaction amount, frequency, location, account behavior, beneficiary relationships, and transaction history before determining whether further action is appropriate. This process can be slow when information is distributed across multiple views or systems.

3.2 Limitations of Existing Systems

• **Excessive false positives:** Fixed thresholds can flag legitimate transactions, increasing analyst workload.

• **Limited adaptability:** Rules must be updated when new risk patterns or behaviors emerge.

• **Weak contextual reasoning:** A rule trigger may not capture the complete behavioral context of a customer or transaction.

• **Manual explanation effort:** Analysts may need to manually document why an alert appears significant or insignificant.

• **High operational effort:** Large alert volumes can increase the time required for investigation and reporting.

• **Fragmented information:** Customer, transaction, beneficiary, location, and historical information may require separate review.

• **Limited natural-language assistance:** Traditional systems generally do not automatically generate analyst-friendly investigation summaries.

These limitations motivate a hybrid architecture in which established compliance controls are complemented by anomaly-oriented analytics and Generative AI-assisted explanation. The proposed system does not assume that rules should be removed; instead, it aims to add analytical and explanatory capabilities around the existing monitoring process.

CHAPTER 4

PROPOSED METHODOLOGY

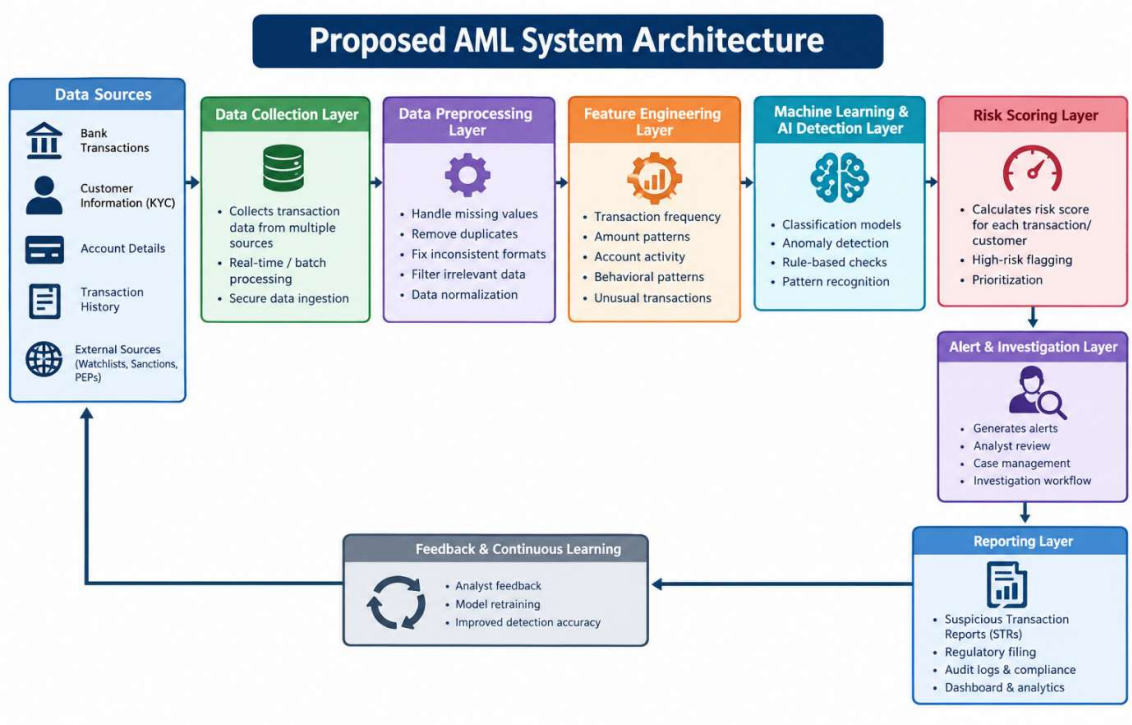
4.1 Overview

The proposed GenAI-Assisted AML Compliance System is a decision-support platform for transaction monitoring and suspicious activity investigation. It combines transaction and behavioral analysis with risk scoring, Generative AI explanations, and a secure investigation dashboard. The system is designed to reduce unnecessary investigation effort while improving the interpretability of alerts. The workflow begins with transaction and customer-related information. Analytical processing considers amount, frequency, location, account behavior, beneficiary relationships, and transaction history. The system then identifies anomalous patterns and assigns customer or transaction risk scores. Alerts can be prioritized so that investigators can focus on cases requiring greater attention. Generative AI receives the relevant analytical context and produces understandable explanations, investigation summaries, contributing risk factors, and compliance-report support. The generated content is displayed to authorized personnel through the investigation dashboard, where the analyst can review the underlying information and validate the case.

4.2 System Architecture

The proposed architecture can be organized into five logical layers: data input, analytical processing, risk and alert management, GenAI explanation, and secure human investigation. The separation of these layers supports maintainability and makes it possible to evolve analytical models or generation components without redesigning the complete interface.

Figure 4.1: Proposed AML system architecture



Description: The architecture connects transaction/customer information to preprocessing and feature analysis, anomaly detection and risk scoring, alert prioritization, Generative AI explanation, and a secure investigation dashboard. Access control, audit logging, data protection, and human validation operate as cross-cutting controls.

4.3 Core Modules

• **Data and transaction intake module:** Collects the transaction, account, beneficiary, location, and historical information required for analysis.

• **Preprocessing and feature module:** Organizes input information into analytical attributes suitable for monitoring and risk assessment.

• **Anomaly detection module:** Identifies unusual patterns using analytical and Machine Learning approaches.

• **Risk scoring module:** Produces customer or transaction risk scores for prioritization.

• **Alert management module:** Creates and prioritizes investigation alerts based on identified risk.

• **GenAI explanation module:** Generates understandable explanations, contributing risk factors, and investigation summaries.

• **Investigation dashboard:** Presents alerts, risk information, investigation status, and generated explanations to authorized users.

• **Security and governance module:** Provides access controls, audit logging, data protection, and support for human validation.

4.4 System Workflow

• **Step 1 – Data intake:** Transaction and related customer information enters the monitoring workflow.

• **Step 2 – Validation and preprocessing:** Input information is checked and prepared for analytical processing.

• **Step 3 – Behavioral analysis:** Transaction amount, frequency, location, account behavior, beneficiary relationships, and history are examined.

• **Step 4 – Anomaly identification:** The analytical layer identifies patterns that differ from expected or observed behavior.

• **Step 5 – Risk scoring:** Customer or transaction risk scores are generated for prioritization.

• **Step 6 – Alert creation:** Higher-priority conditions are presented as investigation alerts.

• **Step 7 – GenAI assistance:** Relevant alert information is summarized and converted into understandable reasons and risk-factor explanations.

• **Step 8 – Analyst review:** An authorized compliance analyst examines the alert, supporting information, and generated explanation.

• **Step 9 – Investigation management:** The analyst updates the investigation status and records relevant actions through the dashboard.

• **Step 10 – Human validation:** The final compliance decision remains under human oversight and is not delegated entirely to the AI system.

4.5 Risk Scoring and Explainability

Risk scoring is intended to prioritize investigation rather than to establish a legal conclusion. A higher score indicates that the available analytical signals warrant greater attention. The system can consider transaction-level and customer-level information so that isolated transactions can be interpreted in a broader behavioral context.

Explainability is provided through natural-language descriptions of the factors associated with an alert. For example, the system can summarize unusual transaction frequency, notable amounts, geographic or location patterns, beneficiary relationships, or differences from historical behavior when those signals are available in the input data. The generated explanation should remain traceable to the underlying information presented to the analyst.

Because Generative AI can produce inaccurate statements, the explanation layer should not be treated as independent evidence. Analysts must validate the generated summary against the original transaction and customer information before relying on it in an investigation or compliance report.

4.6 Advantages and Limitations

Advantages: The proposed system combines analytical monitoring with explainable GenAI assistance, supports risk-based prioritization, reduces manual summarization effort, provides a secure dashboard, and preserves human validation. It also provides a modular foundation in which detection, scoring, explanation, and governance functions can evolve independently.

Limitations: The effectiveness of anomaly detection depends on data quality and analytical design. Risk scores can be affected by incomplete or biased data. Generative AI explanations may be incorrect or incomplete and therefore require validation. Security controls must be correctly implemented to protect sensitive financial information. Finally, the proposed system is a decision-support mechanism and does not by itself guarantee the identification of every laundering pattern or establish legal guilt.

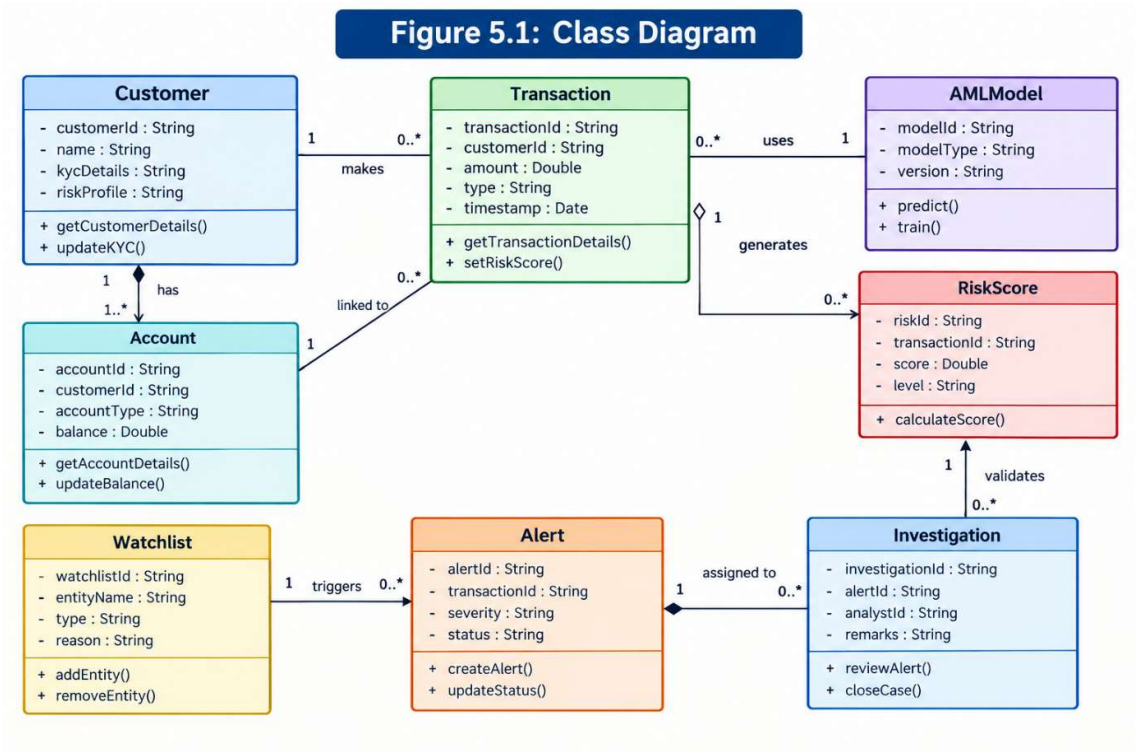
CHAPTER 5

UML DIAGRAMS

5.1 Class Diagram

The class design organizes the major entities required by the proposed AML workflow. Representative classes include Customer, Transaction, Beneficiary, RiskProfile, Alert, Investigation, GenAIExplanation, AuditLog, and AccessController. The Customer and Transaction entities provide core information for analysis, while Alert and Investigation represent the operational review process.

Figure 5.1: Class diagram



Description: The class structure represents customer and transaction data, analytical risk information, alerts, investigations, GenAI-generated explanations, audit records, and access control. Relationships should show that an investigation can contain alerts and generated explanations while audit records capture relevant actions.

5.2 Use Case Diagram

The primary actor is an authorized compliance analyst or investigator. The actor interacts with the system to access alerts, examine transaction and customer risk information, review generated explanations, manage investigation status, and validate outcomes. Administrative functions can additionally control access permissions and review audit information.

Figure 5.2: Use case diagram

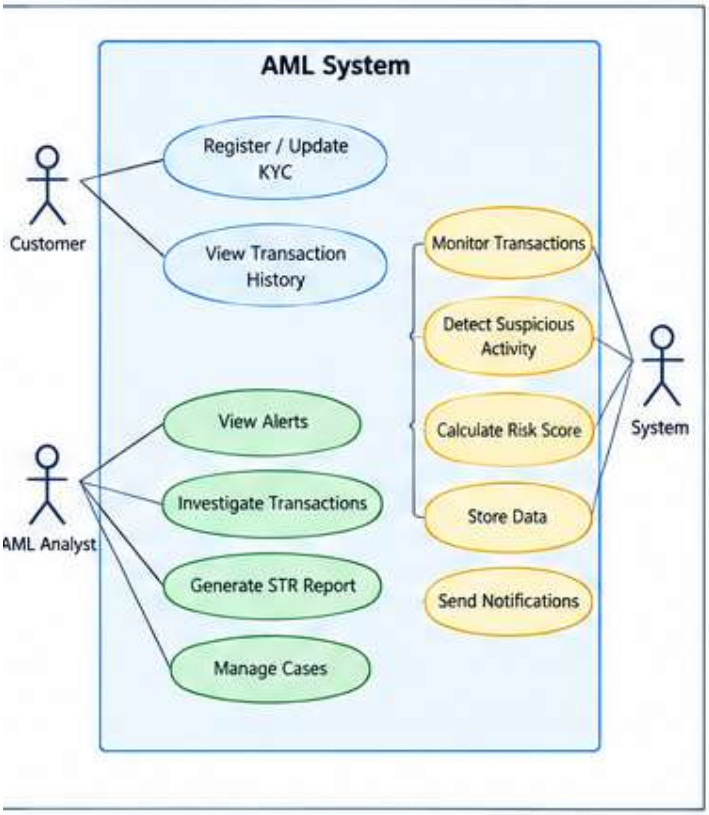


Figure 5.2: Use case diagram

Description: The use cases include secure login/access, viewing alerts, inspecting transaction history, reviewing risk scores, viewing GenAI explanations, updating investigation status, recording analyst decisions, and reviewing audit information.

5.3 Sequence Diagram

The sequence design describes the order in which a transaction-monitoring request is processed. Transaction information is supplied to the monitoring layer, analytical processing

evaluates relevant features, the risk layer generates prioritization information, and the alert is sent to the investigation interface. When explanation assistance is requested, the GenAI component receives the permitted context and returns a natural-language explanation or summary for analyst review.

Figure 5.3: Sequence diagram

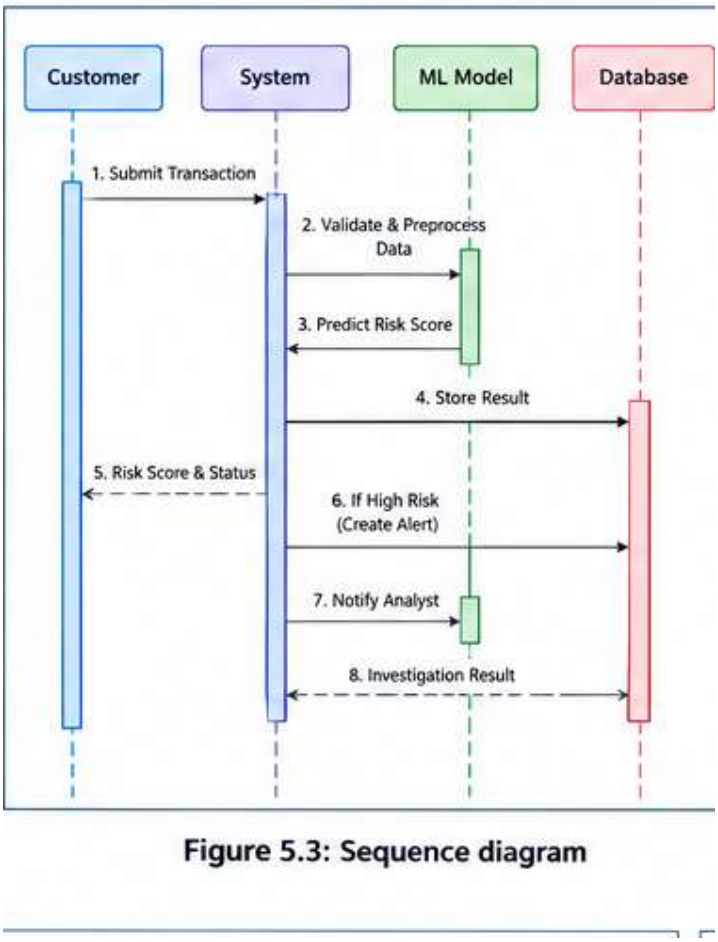


Figure 5.3: Sequence diagram

Description: The sequence should show the client or data source, monitoring service, analytical engine, risk/alert service, GenAI service, dashboard, and analyst. Security and audit events should be represented where appropriate.

5.4 Component Diagram

The component design separates transaction intake, preprocessing, anomaly detection, risk scoring, alert management, GenAI explanation, dashboard presentation, authentication/authorization, and audit logging. This separation supports maintainability and

makes it possible to change the analytical model or GenAI provider without rewriting the complete user interface.

Figure 5.4: Component diagram

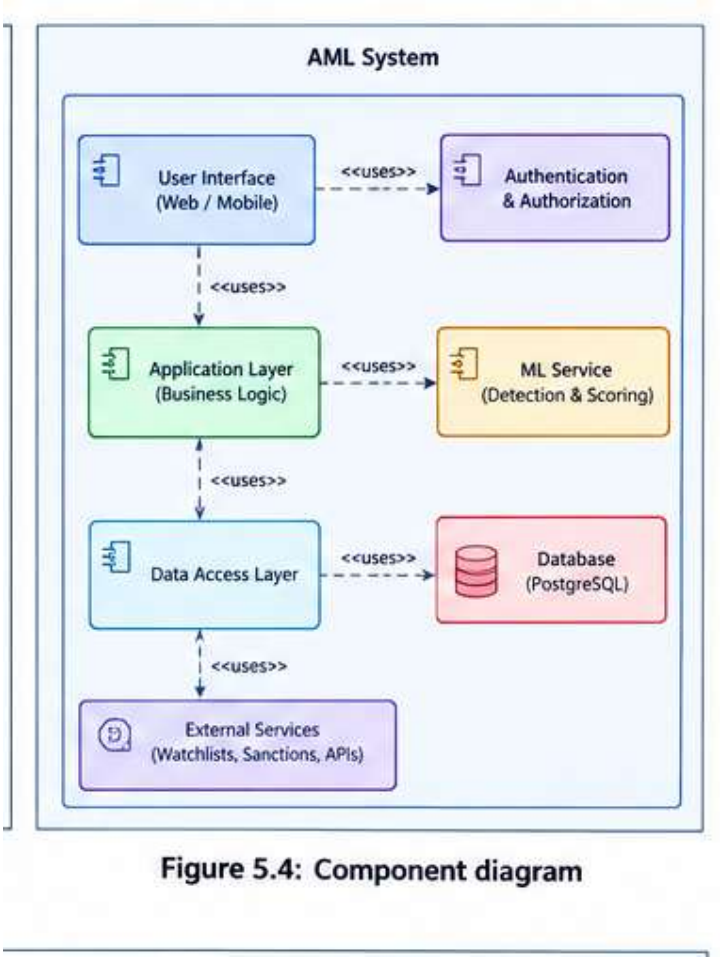


Figure 5.4: Component diagram

Description: The diagram should show modular components and their interfaces, emphasizing separation between analytical processing, GenAI assistance, presentation, and security/governance services.

5.5 Deployment Diagram

The proposed deployment environment contains a client browser, an application server, an analytical processing layer, a secure data store or protected data interface, and a Generative AI service. Communication with external services should use secure network channels. Access to sensitive investigation data should be controlled by authentication and authorization mechanisms.

Figure 5.5: Deployment diagram

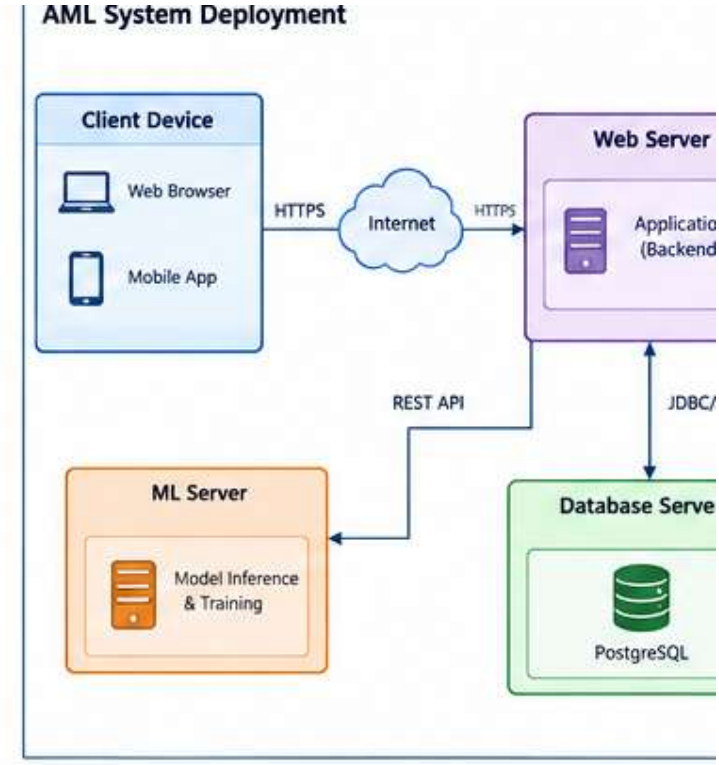


Figure 5.5: Deployment diagram

Description: The deployment view should represent the investigator workstation/browser, application server, protected data layer, analytics services, and external GenAI service connected through controlled network communication.

5.6 Activity Diagram

The activity flow begins with secure access and data intake. The system validates available information, performs preprocessing, analyzes behavioral features, calculates risk information, and creates or updates alerts. If an alert requires investigation, the dashboard presents the case and GenAI assistance can produce an explanation or summary. The analyst reviews the evidence, updates the case, and performs the required human validation.

Figure 5.6: Activity diagram

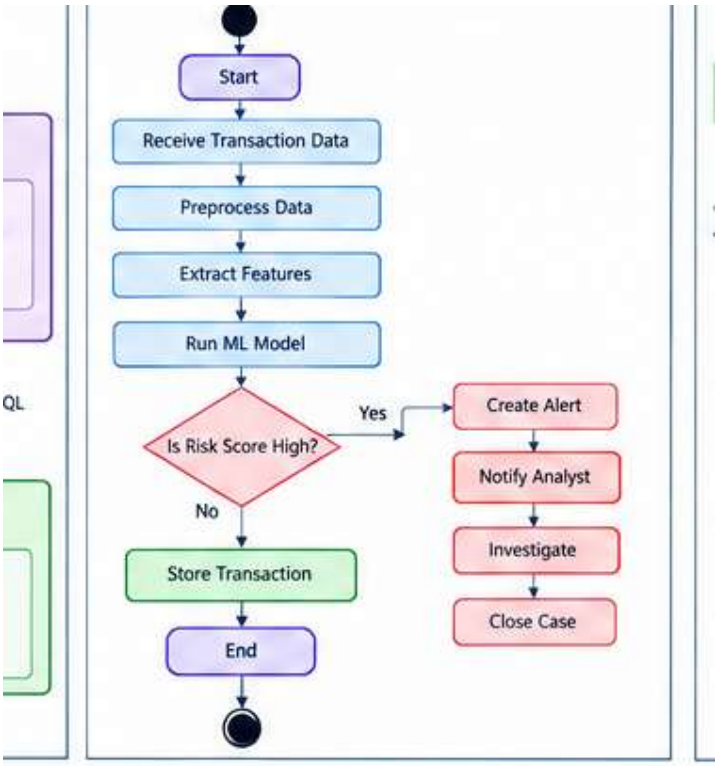


Figure 5.6: Activity diagram

Description: The activity diagram should include authentication, data validation, analysis, anomaly detection, risk scoring, alert prioritization, optional GenAI explanation, analyst review, investigation update, audit logging, and final human validation.

5.7 Data Flow Diagrams (DFD)

5.7.1 DFD Level 0 (Context Level Diagram)

Figure 5.7: DFD Level 0 (context diagram)

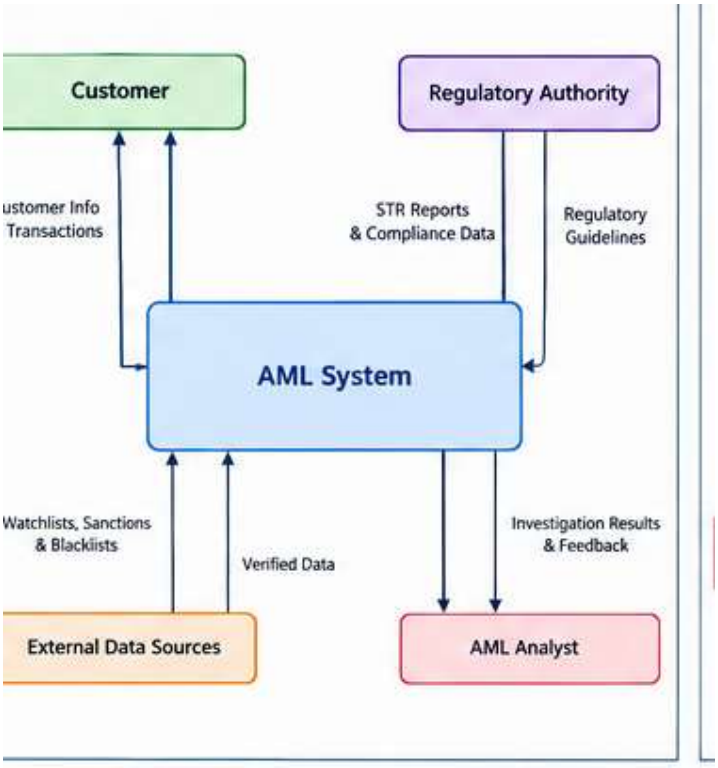


Figure 5.7: DFD Level 0 (context diagram)

Description: External entities include the authorized compliance analyst and relevant transaction/customer data sources. The system receives monitoring information and returns alerts, risk information, explanations, investigation summaries, and status information while maintaining security and audit controls.

5.7.2 DFD Level 1 (Decomposed Process Diagram)

Figure 5.8: DFD Level 1 (decomposed process diagram)

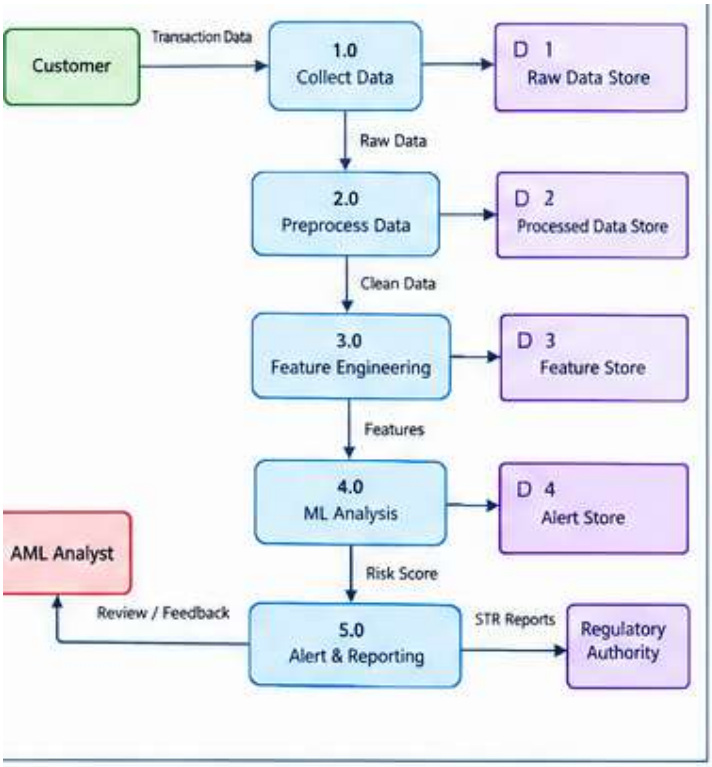


Figure 5.8: DFD Level 1 (decomposed process diagram)

Description: The decomposed flow traces data through validation and preprocessing, feature analysis, anomaly detection, risk scoring, alert generation, GenAI explanation, dashboard presentation, investigation management, and audit logging.

CHAPTER 6
SOFTWARE ENVIRONMENT

6.1 Software Requirements

The following requirements are derived from the proposed architecture and the technologies named in the project abstract. Exact framework versions and implementation dependencies should be finalized according to the project source code before deployment.

• **Operating system:** Windows, Linux, or another supported development/server operating system.

• **Programming language:** Python is suitable for the AI/ML, backend, and orchestration layers.

• **Machine Learning environment:** A Python-based ML stack for anomaly detection and risk-scoring models, with the exact libraries selected during implementation.

• **Natural Language Processing:** NLP components for preparing investigation context and structured information for explanation.

• **Generative AI service:** A supported Generative AI model/service for explanations, investigation summaries, risk-factor descriptions, and report assistance.

• **Web application framework:** A Python web framework such as Flask can provide the secure investigation dashboard and backend API layer.

• **Frontend:** HTML5, CSS3, and JavaScript or an equivalent web interface technology.

• **Data storage:** A protected relational or document-oriented data store for authorized investigation and audit information, depending on implementation.

• **Security/configuration:** Environment-based secret management, authentication, authorization, encrypted communication, and secure configuration.

• **Development tools:** A Python-compatible IDE/editor and version control system such as Git.

6.2 Hardware Requirements

Because the proposed architecture can use external Generative AI services and can separate analytical workloads from the user interface, the end-user hardware requirement can remain modest. Actual production requirements depend on transaction volume, model size,

deployment architecture, and whether ML inference is performed locally or through an external service.

• **Processor:** Modern dual-core or better processor for development and dashboard use; higher capacity is recommended for local analytical workloads.

• **Memory:** At least 8 GB RAM is recommended for development involving Python, analytics, and browser tools; production requirements depend on workload.

• **Storage:** Sufficient storage for source code, datasets, logs, model artifacts, and protected application data.

• **Network:** Reliable internet connectivity is required when the application communicates with external AI services or hosted infrastructure.

• **Client display:** A modern desktop or laptop display suitable for reviewing alerts, transaction history, risk information, and investigation summaries.

CHAPTER 7

SYSTEM REQUIREMENTS SPECIFICATION

The System Requirements Specification defines the functional, technical, hardware, software, user, data, and system requirements of the proposed GenAI-Assisted AML Compliance System.

7.1 Functional Requirements

1. **User Authentication:** Allows authorized users to securely log in and access the AML system.
2. **Transaction Data Processing:** Collects and processes transaction information for AML analysis.
3. **Transaction Monitoring:** Monitors transactions and identifies unusual or suspicious patterns.
4. **Anomaly Detection:** Uses Machine Learning and AI techniques to detect anomalous transaction behavior.
5. **Risk Scoring:** Assigns risk scores to customers or transactions based on analyzed factors.
6. **Alert Generation:** Generates alerts for potentially suspicious or high-risk activities.
7. **Alert Explanation:** Provides understandable reasons and risk factors behind generated alerts.
8. **Investigation Support:** Enables analysts to review alerts, transaction details, risk information, and investigation status.
9. **Generative AI Assistance:** Uses Generative AI to summarize investigations and assist in preparing compliance reports.
10. **Case Prioritization:** Helps analysts prioritize high-risk cases for further investigation.
11. **Dashboard:** Provides a secure dashboard for viewing alerts, risks, and investigation status.
12. **Audit Logging:** Maintains records of user and investigation activities for auditing.
13. **Human Validation:** Supports human review and validation before final compliance decisions.

7.2 Non-Functional Requirements

1. **Security:** Protects customer and transaction information through access controls and data protection.

2. **Performance:** Processes transactions efficiently and provides alerts within an acceptable response time.
3. **Reliability:** Provides consistent results and prevents loss of important investigation information.
4. **Scalability:** Handles increasing customers, transactions, alerts, and investigation records efficiently.
5. **Usability:** Provides a simple and understandable interface for compliance analysts.
6. **Explainability:** Provides clear explanations for AI-generated alerts and risk assessments.
7. **Maintainability:** Uses a modular design that allows components to be easily updated and maintained.